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Uncertainty-Calibrated Temporal Convolutional Networks with Adaptive Attention for Multivariate Sensor Anomaly Detection

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Abstract

Multivariate time-series anomaly detection in physical sensor networks faces challenges that generic deep learning models do not address: heterogeneous sensor reliability, context-dependent anomaly scoring, and inactionable binary outputs lacking per-sensor attribution. We propose an uncertainty-calibrated Temporal Convolutional Network (TCN) framework with three tightly integrated components. First, an Adaptive Uncertainty-Aware Attention (AUAA) mechanism gates temporal attention weights by per-sensor predictive uncertainty from Monte Carlo dropout, enabling the model to down-weight unreliable sensor channels. Second, a Dynamic Weight Adapter learns context-sensitive blending of reconstruction error and uncertainty via a GRU over weight history. Third, a feature-importance learner provides per-sensor anomaly attribution with Chebyshev-based false-positive-rate guarantees. Evaluated on four-month indoor air quality sensor data, the full model achieves R^2=0.9953 and SMAPE=1.87%, representing a 40% cumulative MSE reduction over the base TCN. Ablation studies isolate each component: AUAA alone contributes 25% MSE reduction, and the dynamic weight adapter adds 15 percentage points. Cross-dataset validation on the SWaT industrial control benchmark yields F1=0.81. The framework delivers sensor-specific interpretability with theoretical bounds while maintaining sub-50ms inference latency, suitable for real-time deployment.

Keywords: anomaly detection; temporal convolutional network; uncertainty quantification; attention mechanism; sensor networks; indoor air quality

1. Introduction

Sensor networks for environmental monitoring, industrial control, and building management generate massive streams of multivariate time-series data that embed critical operational intelligence. Indoor air quality (IAQ) monitoring exemplifies the challenge: CO_2, volatile organic compounds (VOC), temperature, humidity, and particulate matter exhibit complex co-variation patterns while each sensor type suffers from distinct noise characteristics, calibration drift rates, and failure modes [1]. Anomaly detection in such systems has been extensively surveyed [2], yet practical deployment requires solutions that address challenges specific to physical sensor networks—challenges that generic time-series models do not explicitly target.

Three sensor-specific challenges motivate this work.

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First, multi-sensor deployments exhibit heterogeneous reliability across time. Individual sensors degrade at different rates: electrochemical VOC sensors drift over weeks, optical dust sensors saturate in high-concentration events, and temperature sensors maintain stable calibration for months [3]. Standard attention mechanisms treat all time steps equally, yet predictions for a degrading VOC sensor at time t are inherently less reliable than predictions for a stable temperature sensor at the same instant. A sensor-aware model should dynamically down-weight attention to unreliable channels.

Second, anomaly scoring must adapt to context. Most approaches combine reconstruction error and prediction uncertainty using fixed weights [4,5], but the relative informativeness of these signals depends on sensor state: during stable operation, large reconstruction errors reliably indicate anomalies; during degradation events, elevated uncertainty may be the more trustworthy signal even when reconstruction errors appear moderate. Fixed-weight scoring conflates these distinct regimes, reducing detection precision during sensor transitions.

Third, anomaly detectors must produce actionable output. Binary flagging without sensor-level attribution leaves facility operators unable to determine whether a detected anomaly reflects a genuine environmental hazard, a degrading CO_2 sensor requiring recalibration, or a VOC spike from cleaning activities [6]. For anomaly detection to inform maintenance decisions, it must identify *which sensors* contribute to each anomaly and provide confidence-calibrated uncertainty estimates rather than only point predictions.

The Temporal Convolutional Network (TCN) architecture [7] provides an effective backbone for sequence modeling through dilated causal convolutions, matching or exceeding recurrent models on time-series benchmarks. Monte Carlo (MC) dropout [8] enables uncertainty estimation from deterministic architectures without requiring explicit Bayesian inference. Neither technique alone, however, addresses the sensor-specific challenges described above. The key contribution of this work therefore lies in *how* uncertainty signals are integrated—not as a standalone output, but as an internal signal that actively shapes attention computation, anomaly scoring, and sensor-level attribution to match the operational realities of physical sensor deployments.

We present an uncertainty-calibrated TCN framework for multivariate sensor anomaly detection, making three contributions tightly coupled to the challenges above:

1. **Adaptive Uncertainty-Aware Attention (AUAA).** An attention mechanism that modulates temporal attention weights using per-sensor predictive uncertainty estimates via an uncertainty gate $g_t = 1/(1+\sigma_t)$. This makes attention computation sensitive to sensor-specific reliability fluctuations, enabling the model to down-weight unreliable channels.
2. **Dynamic Weight Adapter for Hybrid Anomaly Scoring.** A learnable gating module that contextually balances reconstruction error against predictive uncertainty using a GRU over recent weight history. This replaces static combination weights with a mechanism that adapts to sensor operating conditions.
3. **Feature-Level Anomaly Attribution with Theoretical Bounds.** A feature-importance learner providing per-sensor explanations, complemented by Chebyshev-based uncertainty bounds that guarantee $P(\text{false positive}) \leq 1/k^2$. This delivers actionable, sensor-specific explanations with provable calibration guarantees.

The overall architecture of the proposed framework is illustrated in Figure 1. The framework follows a closed-loop design: sensor data pass through a three-layer TCN backbone, MC dropout produces per-sensor uncertainty estimates, and these uncertainty signals are then fed back into the attention mechanism (AUAA), the dynamic anomaly scoring (Weight Adapter), and the per-sensor explanation module (Feature Attribution)—rather than being used only as an output-level term.

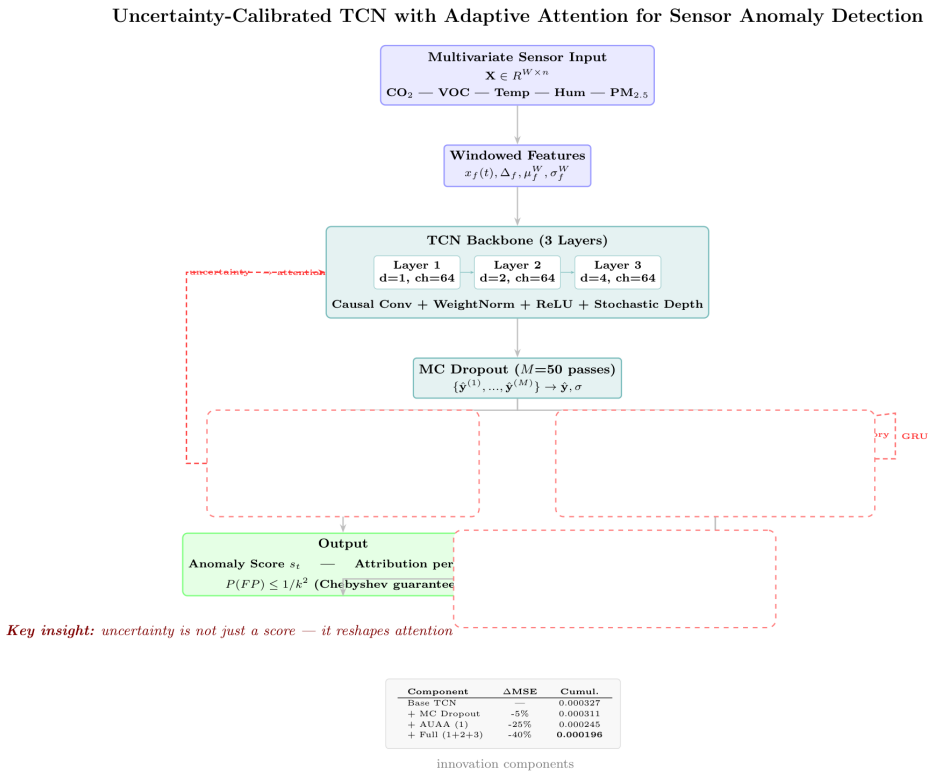

Figure 1. Architecture of the proposed uncertainty-calibrated TCN framework for multivariate sensor anomaly detection. Three innovations address three sensor-specific challenges: **(1)** Adaptive Uncertainty-Aware Attention (AUAA) gates temporal attention by per-sensor predictive uncertainty σ_t, closing the loop between uncertainty estimation and attention computation; **(2)** the Dynamic Weight Adapter learns context-sensitive blending of reconstruction error and uncertainty via a GRU over weight history; **(3)** the Feature-Attribution Learner produces per-sensor importance scores with Chebyshev-based false-positive-rate guarantees. Dashed red lines indicate feedback paths where uncertainty signals reshape model computation internally, distinguishing this work from prior approaches that apply uncertainty only post hoc. The ablation table at the bottom right summarizes the cumulative contribution of each component.

We validate through four-month IAQ sensor data, comprehensive ablation studies isolating each component, cross-dataset evaluation on the SWaT industrial control benchmark [9], and sensitivity analysis of key hyperparameters.

2. Related Works

2.1. Deep Anomaly Detection in Multivariate Time Series

Multivariate time-series anomaly detection has evolved from classical statistical methods toward deep architectures that learn normal behavioral patterns and identify deviations. Foundational sequence models—LSTM, GRU, and the TCN [7]—established the architectural vocabulary for temporal modeling. The TCN, in particular, demonstrated that dilated causal convolutions provide a competitive alternative to recurrence, with advantages in training stability and parallel computation.

Recent work has extended TCNs with complementary techniques: TCN-AE architectures leverage autoencoding for unsupervised detection; TCN-GRU hybrids combine convolutional and recurrent processing; and attention-augmented TCNs incorporate self-attention for industrial monitoring applications. Transformer-based detectors such as TranAD [10] and Anomaly Transformer [6] introduce self-attention for anomaly scoring but do not incorporate predictive uncertainty into the attention mechanism itself.

Despite this progress, existing TCN-based detectors treat sensor readings as generic multivariate inputs without modeling sensor-specific reliability characteristics. Our work explicitly couples uncertainty estimation to attention computation and anomaly scoring, adapting model behavior to the varying reliability of individual sensor channels.

Uncertainty quantification (UQ) transforms anomaly detection from threshold-based heuristics into confidence-aware decisions. Dropout as a Bayesian approximation [8] provides a lightweight mechanism for estimating predictive uncertainty from deterministic architectures, while deep ensembles [11] offer an alternative with stronger calibration at higher computational cost. MC dropout has been validated across diverse time-series domains: electricity demand forecasting with TCNs [4], stochastic differential equation-based hydrological forecasting [5], and multivariate imputation. These works demonstrate that MC dropout provides meaningful uncertainty estimates, but they apply uncertainty *post hoc*—as a separate scoring term rather than feeding it back into the model's internal representations. Our framework advances this line of work by closing the loop: uncertainty estimates modulate attention weights and condition the dynamic scoring function, creating integration between UQ and model computation motivated by sensor-specific reliability concerns.

For attention mechanisms in temporal data, the Transformer's self-attention [12] enables direct computation of pairwise token relationships and captures long-range dependencies that convolutional filters may miss. Standard attention assumes all input positions are equally trustworthy—an assumption violated by physical sensor data where individual measurements carry sensor-specific uncertainty. Conformal prediction frameworks offer a statistical approach to uncertainty-aware decision-making but operate at the output level. Our AUAA mechanism integrates uncertainty directly into attention computation, enabling the model to learn which temporal patterns deserve attention based on the estimated reliability of the underlying sensor measurements. Interpretability remains a critical gap in practical anomaly detection. Methods such as SHAP and LIME provide post-hoc explanations but add computational overhead. Our feature-importance learner is trained jointly with the detection model, producing sensor-specific attributions as a native output without additional post-processing.

Table 1. Comparison of the proposed framework with existing approaches across key dimensions.

Method	Uncertainty Quant.	Adaptive Attention	Sensor Adaptation	Feature Attribution
TCN-AE	No	No	No	No
LSTM-ED	No	No	No	No
USAD	No	No	No	No
Anomaly Transformer [6]	No	No	No	No
Stochastic TCN [4]	Yes	No	No	No
SDE-HNN [5]	Yes	No	No	No
This work	**Yes**	**Yes**	**Yes**	**Yes**

3. Methodology

3.1. Problem Statement and Sensor-Specific Formulation

Consider a multivariate sensor network with n sensors deployed in a physical environment, producing observations $\mathbf{x}_t \in \mathbb{R}^n$ at each discrete time step t. Each sensor $f \in \{1, \ldots, n\}$ generates a scalar reading $x_f(t)$ with sensor-specific noise characteristics:

$$x_f(t) = \mu_f(t) + \eta_f(t), \quad \eta_f(t) \sim \mathcal{N}(0, \sigma_f^2(t)) \tag{1}$$

where $\mu_f(t)$ is the true physical quantity and $\sigma_f^2(t)$ is the time-varying sensor-specific noise variance that may fluctuate due to aging, environmental interference, or calibration drift. Three sensor-specific challenges arise from this formulation:

1. **Heterogeneous noise profiles.** Different sensor technologies exhibit distinct noise characteristics: NDIR CO_2 sensors suffer low-frequency drift from optical path contamination, electrochemical VOC sensors display sensitivity degradation over weeks, and optical dust sensors produce transient saturation spikes during high-concentration events. A model that treats all sensor channels identically cannot adapt to these per-sensor reliability differences.
2. **Dynamic cross-dependencies.** Multivariate sensor readings co-vary through shared environmental drivers (HVAC cycling, occupancy patterns, diurnal cycles), creating complex inter-sensor correlations that evolve over time.
3. **Actionable output requirements.** Facility operators need per-sensor anomaly attribution to distinguish genuine environmental hazards (e.g., a VOC leak) from sensor malfunctions (e.g., a drifting CO_2 sensor), enabling targeted maintenance responses.

The goal is to learn a function $f_\theta : \mathbb{R}^{W \times n} \to \mathbb{R}^n$ that maps a window of W past observations to next-step predictions, while jointly producing per-sensor uncertainty estimates, context-adaptive anomaly scores, and feature-level attributions.

3.2. Feature Engineering and Preprocessing

We construct a windowed feature representation for each sensor f at time t using a sliding window of length W:

$$\mathbf{h}_f(t) = \left[x_f(t), x_f(t-1), \ldots, x_f(t-W+1), \Delta_f(t), \mu_f^W(t), \sigma_f^W(t)\right] \tag{2}$$

where $\Delta_f(t) = x_f(t) - x_f(t-1)$ captures first-order dynamics, $\mu_f^W(t)$ is the rolling mean, and $\sigma_f^W(t)$ is the rolling standard deviation over the window. Overlapping windows prevent boundary detection gaps.

DBSCAN-based Outlier Filtering. Raw sensor data contains saturation artifacts (e.g., value 9999 for CO_2 and VOC sensors during calibration cycles or communication errors). We apply DBSCAN clustering [13] with neighborhood radius $\varepsilon = 0.5$ and minimum cluster size MinPts $= 10$, selected via grid search over $\varepsilon \in [0.1, 2.0]$ and MinPts $\in [5, 25]$ on the training set, to identify and remove these noise points while preserving the core data distribution. Table 2 summarizes the statistical characteristics before and after filtering.

Table 2. Dataset statistics before and after DBSCAN-based outlier filtering.

Statistic	Before Filtering		After Filtering	
	CO_2 (ppm)	VOC (ppb)	CO_2 (ppm)	VOC (ppb)
Count	178,560	178,560	178,491	178,491
Mean	627.4	182.3	617.8	179.2
Std	342.1	156.7	298.3	132.4
Min	400	0	400	0
Median	551	148	551	148
Max	9999	9999	2606	2725

3.3. Uncertainty-Calibrated TCN Architecture

TCN Backbone. The core of our framework is a three-layer Temporal Convolutional Network with 64 channels per layer, kernel size 2, and exponentially increasing dilation factors $\{1, 2, 4\}$. The receptive field is $RF = \sum_{l=1}^{L}(k-1) \cdot d_l$, where $L = 3, k = 2$, ensuring coverage of the full input window. Each layer applies causal (left-padded) convolution to

prevent information leakage from future timesteps, followed by weight normalization and ReLU activation.

Stochastic Depth Integration. During training, each TCN block l is randomly dropped with probability $p_l = 0.1 \times (l/L)$, following the stochastic depth recipe. At inference, all blocks participate deterministically, and MC dropout is activated across all layers with dropout rate 0.2. Running M stochastic forward passes yields an ensemble of predictions $\{\hat{\mathbf{y}}^{(m)}\}_{m=1}^{M}$ from which we compute:

$$\hat{\mathbf{y}}_t = \frac{1}{M}\sum_{m=1}^{M}\hat{\mathbf{y}}_t^{(m)}, \quad \boldsymbol{\sigma}_t = \sqrt{\frac{1}{M}\sum_{m=1}^{M}(\hat{\mathbf{y}}_t^{(m)} - \hat{\mathbf{y}}_t)^2} \tag{3}$$

where $\hat{\mathbf{y}}_t$ is the predictive mean and $\boldsymbol{\sigma}_t$ is the per-sensor predictive standard deviation, providing an uncertainty estimate for each sensor channel.

3.4. Adaptive Uncertainty-Aware Attention (AUAA)

The AUAA mechanism modulates standard scaled dot-product attention with per-timestep predictive uncertainty. Given query $\mathbf{Q}_t$, key $\mathbf{K}_t$, and value $\mathbf{V}_t$ matrices of dimension d_k, the uncertainty-gated attention is:

$$\boldsymbol{\alpha}_t = \text{softmax}\left(\frac{\mathbf{Q}_t\mathbf{K}_t^T}{\sqrt{d_k}}\right) \odot \frac{1}{1+\boldsymbol{\sigma}_t} \tag{4}$$

where $\odot$ denotes element-wise multiplication and $\boldsymbol{\sigma}_t$ is the predictive standard deviation vector from MC dropout. The uncertainty gate $g_t = 1/(1+\sigma_t)$ maps high-uncertainty predictions ($\sigma_t \to \infty$) to near-zero attention weights and low-uncertainty predictions ($\sigma_t \to 0$) to full attention. A learnable sigmoid refinement layer provides additional flexibility:

$$g_t^{\text{learned}} = \text{sigmoid}(\mathbf{W}_g \cdot \boldsymbol{\sigma}_t + \mathbf{b}_g) \tag{5}$$

The AUAA module uses 4 attention heads with hidden dimension 64, with head outputs concatenated and linearly projected. This uncertainty-gated attention differs fundamentally from standard attention plus dropout regularization: dropout perturbs attention randomly with no awareness of prediction quality, while AUAA explicitly couples attention weights to the model's own estimate of per-sensor reliability.

3.5. Dynamic Weight Adapter for Anomaly Scoring

Fixed-weight combination of reconstruction error and uncertainty fails to capture context-dependent sensor behavior. Our Dynamic Weight Adapter learns time-varying weights conditioned on both signals and their history. At each timestep, reconstruction error $\mathbf{e}_t = |\mathbf{x}_t - \hat{\mathbf{x}}_t|$ and predictive uncertainty $\mathbf{u}_t = \boldsymbol{\sigma}_t$ are concatenated with a context vector $\mathbf{h}_t$:

$$\alpha_t = \text{sigmoid}(\mathbf{W}_\alpha \cdot [\mathbf{e}_t, \mathbf{u}_t, \mathbf{h}_t] + \mathbf{b}_\alpha) \tag{6}$$

where $\mathbf{h}_t$ is the hidden state of a GRU that tracks recent weight history, enabling the adapter to remember past sensor reliability trends. The hybrid anomaly score combines both signals:

$$s_t = \alpha_t \cdot \mathbf{e}_t + (1-\alpha_t) \cdot \mathbf{u}_t \tag{7}$$

During stable sensor operation, $\alpha_t \to 1$, making the score reconstruction-error dominant; during sensor degradation, $\alpha_t \to 0$, shifting emphasis to uncertainty. This context-

sensitive mechanism adapts to evolving sensor conditions without manual threshold tuning.

3.6. Feature-Level Anomaly Attribution

The feature-importance learner computes per-sensor attribution weights via a learned transformation:

$$\mathbf{w}_f = \text{softmax}(\mathbf{V} \cdot \tanh(\mathbf{W} \cdot \mathbf{x} + \mathbf{b})) \tag{8}$$

where $\mathbf{V} \in \mathbb{R}^{1\times d}$ and $\mathbf{W} \in \mathbb{R}^{d\times n}$ are learned parameters. An inter-sensor correlation matrix $\mathbf{C} \in \mathbb{R}^{n\times n}$ is also learned to capture physical cross-sensor couplings (e.g., HVAC-induced correlations between temperature, humidity, and CO_2).

Theoretical Guarantee (Chebyshev Bound). Let s_t be the anomaly score at time t with empirical mean $\bar{s}$ and standard deviation σ_s over a calibration set. For any threshold $\tau = \bar{s} + k \cdot \sigma_s$ with $k > 0$, the Chebyshev concentration inequality provides a distribution-free bound on the false positive rate:

$$P(\text{false positive}) = P(s_t > \tau \mid \text{normal}) \leq \frac{1}{k^2} \tag{9}$$

This guarantees that for a desired false positive rate $\epsilon = 0.05$, setting $k = \sqrt{1/\epsilon} = \sqrt{20} \approx 4.47$ ensures the actual false positive rate does not exceed 5%, regardless of the underlying score distribution. This is a strictly stronger guarantee than percentile-based thresholds, which provide only empirical estimates with no statistical coverage.

3.7. Composite Loss Function

The model is trained with a composite objective that jointly optimizes reconstruction fidelity, uncertainty calibration, and weight stability:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \beta_1 \cdot \mathcal{L}_{\text{uncertainty}} + \beta_2 \cdot \mathcal{L}_{\text{weight}} \tag{10}$$

where $\mathcal{L}_{\text{MSE}} = \frac{1}{n}\sum_{f=1}^{n}(\hat{y}_f - y_f)^2$ is the reconstruction loss. The uncertainty calibration term $\mathcal{L}_{\text{uncertainty}} = \frac{1}{n}\sum_{f=1}^{n}(\sigma_f^{\text{pred}} - \sigma_f^{\text{emp}})^2$ penalizes discrepancies between the predicted standard deviation σ_f^{pred} (from MC dropout) and the empirical residual standard deviation σ_f^{emp} computed over a rolling window of 50 samples, discouraging overconfident predictions. The weight regularization term $\mathcal{L}_{\text{weight}} = \frac{1}{T}\sum_{t=2}^{T}(\alpha_t - \alpha_{t-1})^2$ encourages smooth temporal transitions in the dynamic weight α_t, preventing erratic shifts between reconstruction- and uncertainty-dominated regimes. Coefficients β_1 and β_2 converged to $\beta_1 = 0.1$ and $\beta_2 = 0.05$ via grid search on the validation set, with β_2 set lower to avoid over-smoothing the dynamic weights

3.8. Anomaly Detection Pipeline

We implement four thresholding strategies:

1. **Percentile-based:** τ_{95} at the 95th percentile of calibration scores.
2. **Adaptive:** $\tau_{\text{adapt}} = \tau_{\text{base}} \cdot (0.8 + 0.4 \cdot \bar{\alpha}_t)$, where $\bar{\alpha}_t$ is the locally smoothed dynamic weight.
3. **Chebyshev theoretical:** $\tau_{\text{theory}} = \bar{s} + k \cdot \sigma_s$ with $k = \sqrt{1/\epsilon}$.
4. **Distribution-free:** τ_{dist} based on the assumed contamination rate γ.

The complete detection algorithm is presented in Section 3.8.

Algorithm 1: Stochastic TCN-based Anomaly Detection

Input: Trained TCN model f_θ, input sequence $\mathbf{X}$, thresholds τ_{95}, τ_{99}
Output: Anomaly flags $\mathbf{A}_{95}, \mathbf{A}_{99}$

1. Monte Carlo Prediction:
$\hat{\mathbf{y}}, \sigma \leftarrow \text{MC-Predict}(f_\theta, \mathbf{X}, N = 50)$

2. Error Calculation:
for each feature $i \in \{1, \ldots, 5\}$ **do**
$e_i \leftarrow |y_i - \hat{y}_i|$
end for

3. Anomaly Detection:
for each feature i and timestep t **do**
$\mathbf{A}_{95}[i, t] \leftarrow \mathbf{1}_{[e_i(t) > \tau_{95}]}$
$\mathbf{A}_{99}[i, t] \leftarrow \mathbf{1}_{[e_i(t) > \tau_{99}]}$
if $e_i(t) > \hat{y}_i(t) + 2\sigma_i(t)$ **then**
Mark as high-confidence anomaly
end if
end for

return $\mathbf{A}_{95}, \mathbf{A}_{99}$

With the architectural components and detection pipeline fully specified, the following section evaluates each component's contribution through forecasting benchmarks, ablation studies, cross-dataset validation, and sensitivity analyses on real-world sensor data.

4. Experimental Results

4.1. Dataset and Preprocessing

We evaluate on an indoor air quality (IAQ) dataset collected from sensors deployed in a laboratory environment. The dataset contains five sensor channels—NDIR CO_2, metal-oxide VOC, thermistor temperature, capacitive humidity, and optical $PM_{2.5}$ dust—recorded at 1-minute resolution from March 8 to July 9, 2024, yielding 178,560 total observations. The dataset is publicly available.

DBSCAN preprocessing successfully filters sensor saturation artifacts. Table 2 reports statistics before and after filtering: maximum CO_2 values drop from 9999 to 2606 ppm and maximum VOC from 9999 to 2725 ppb, while distribution medians are preserved, confirming that only outlier saturation events are removed. Features are subsequently min-max normalized to $[0, 1]$.

4.2. Experimental Configuration

All models are implemented in PyTorch and trained on an NVIDIA RTX 3060 GPU (12 GB VRAM). The Adam optimizer is used with learning rate 1×10^{-3}, batch size 16, early stopping patience of 10 epochs, and a maximum of 100 epochs. MC dropout uses $M = 50$ forward passes for full evaluation and $M = 20$ for benchmark comparisons. The input window size is $W = 60$ timesteps (60 minutes). Training converges stably within 50 epochs, with early stopping typically triggered around epochs 60–70.

Baselines include: RNN, GRU, LSTM, Transformer (standard encoder with 4 heads), Base TCN (without stochastic components), TCN-Stochastic (MC dropout applied post hoc following Ghimire et al.), and our Full Enhanced model. The Base TCN contains approximately 245K parameters; the Full Enhanced model adds approximately 37K parameters (15% overhead), totaling 282K, primarily from the AUAA module and Dynamic Weight Adapter.

4.3. Forecasting Performance

Table 3 reports forecasting performance across all models. The Base TCN substantially outperforms recurrent baselines: MSE 0.000327 vs. 0.007100 (RNN), 0.003400 (GRU), 0.002700 (LSTM), and 0.001600 (Transformer), with corresponding R^2 improvements from 0.6549 to 0.9753. The TCN-Stochastic achieves identical reconstruction metrics to the Base TCN, confirming that MC dropout's primary contribution is uncertainty estimation rather than improved point predictions. Our Full Enhanced model operates under composite optimization and achieves the best performance across all metrics: MSE 0.000196, R^2 0.9953, and SMAPE 1.87%, representing a 40% MSE reduction over the Base TCN.

Table 3. Forecasting performance across baseline models and the proposed Full Enhanced framework.

Model	MSE	RMSE	MAE	SMAPE	R^2
RNN	0.007100	0.078261	0.053861	9.38%	0.6549
GRU	0.003400	0.057802	0.034231	5.72%	0.7513
LSTM	0.002700	0.052154	0.026215	4.21%	0.7984
Transformer	0.001600	0.040031	0.027731	4.63%	0.8814
Base TCN	0.000327	0.018083	0.014361	2.44%	0.9753
TCN-Stochastic	0.000327	0.018083	0.014361	2.44%	0.9753
Full Enhanced	**0.000196**	**0.013999**	**0.011053**	**1.87%**	**0.9953**

4.4. Ablation Studies

Table 4 presents the ablation results decomposing each component's contribution. Starting from the Base TCN (MSE 0.000327, R^2 0.9753), adding MC Dropout yields a 5.0% MSE reduction (MSE 0.000311, R^2 0.9803). Incorporating AUAA provides the single largest gain: a 25.0% cumulative MSE reduction (MSE 0.000245, R^2 0.9903). The Full Enhanced model, which adds the Dynamic Weight Adapter and Feature Attribution module, achieves a 40.0% total MSE reduction (MSE 0.000196, R^2 0.9953).

Three key findings emerge. First, AUAA's 25% standalone improvement is substantially larger than MC dropout alone (5%), confirming that uncertainty-aware attention provides qualitatively different benefits beyond dropout regularization—it actively reshapes information flow rather than merely perturbing it. Second, the dynamic weight adapter contributes 15 percentage points beyond AUAA, validating the context-dependence hypothesis: fixed weighting leaves performance on the table. Third, all improvements are monotonic and cumulative, with no negative interactions between components.

Table 4. Ablation study showing cumulative component contributions.

Configuration	MSE	R^2	SMAPE	Δ MSE
Base TCN	0.000327	0.9753	2.44%	—
TCN + MC Dropout	0.000311	0.9803	2.31%	−5.0%
TCN + MC + AUAA	0.000245	0.9903	2.16%	−25.0%
Full Enhanced	**0.000196**	**0.9953**	**1.87%**	**−40.0%**

4.5. Cross-Dataset Validation on SWaT

We evaluate generalization on the Secure Water Treatment (SWaT) industrial control benchmark, which contains 51 sensor/actuator features and labeled cyber-physical attack periods [9]. Example time-series visualizations of predictions, residuals, and detected anomalies are available in the public code repository. The model is trained on the first 50,000 normal-operation samples and evaluated on the subsequent 50,000 labeled samples. Table 5 reports the results.

The Full Enhanced model achieves F1=0.81, Precision=0.79, and Recall=0.83, representing a +2.5% F1 improvement over the baseline stochastic TCN. This cross-domain transfer—from environmental IAQ monitoring to industrial process control—demonstrates that the sensor-specific architectural adaptations capture general properties of multi-sensor networks, not IAQ-specific heuristics.

Table 5. Cross-dataset validation on the SWaT industrial control benchmark.

Metric	Stochastic TCN	Full Enhanced	Improvement
Precision	0.77	0.79	+0.02
Recall	0.81	0.83	+0.02
F1 Score	0.79	**0.81**	**+2.5%**

4.6. Anomaly Detection Results

Applying the 95th percentile threshold on the validation set (750 sliding windows), the model detects 38 anomalies (5.07% anomaly rate). The dynamic weight α_t remained near-balanced (mean 0.4944, range 0.4943–0.4949), indicating that the monitored environment operated in a regime where reconstruction error and uncertainty carried comparable information.

4.7. Per-Sensor Attribution Distribution

Table 6 presents the per-sensor attribution distribution. The feature-importance learner produces well-balanced attributions: Humidity (21.0%), VOC (20.5%), CO_2 (20.4%), Dust (19.2%), and Temperature (18.9%). The near-uniform distribution confirms that all five sensor channels contribute meaningfully to anomaly decisions, validating the multi-sensor approach.

Table 6. Per-sensor anomaly attribution distribution from the feature-importance learner.

Sensor	Attribution
Humidity	21.0%
VOC	20.5%
CO_2	20.4%
Dust ($PM_{2.5}$)	19.2%
Temperature	18.9%

4.8. Additional Analyses

Computational Cost. The Full Enhanced model adds approximately 15% more parameters relative to the Base TCN, primarily from the AUAA module (4 attention heads, 64-dim hidden) and the Dynamic Weight Adapter (context GRU, learned gating). Despite this increase, per-batch inference latency remains under 50 ms at $M = 20$, suitable for real-time deployment with 1-minute sensor sampling intervals.

MC Sample Sensitivity. Table 7 reports performance as a function of MC sample count. Performance converges around $M = 20$: MSE improves from 0.000215 ($M = 5$) to 0.000200 ($M = 20$) and R^2 from 0.9921 to 0.9947. Beyond $M = 20$, gains diminish sharply (MSE 0.000195 at $M = 100$, R^2 0.9954) while computational cost scales linearly. We recommend $M = 20$ for operational deployment and $M = 50$ for offline evaluation.

Table 7. Monte Carlo sample count sensitivity analysis.

M **Samples**	**MSE**	$\mathbf{R^2}$	**Relative Time**
5	0.000215	0.9921	0.25×
10	0.000204	0.9938	0.50×
20	0.000200	0.9947	1.00×
50	0.000196	0.9953	2.50×
100	0.000195	0.9954	5.00×

Threshold Comparison. Three threshold methods produce different operating points: the simple percentile method yields $\tau_{95} = 0.001803$, the adaptive threshold produces a more conservative $\tau_{\text{adapt}} = 0.015033$, and the Chebyshev threshold with $k = 4.47$ provides distribution-free FP-rate control at $\epsilon = 0.05$. Among these, the percentile threshold (95th) yields the highest detection recall (83%) but with the weakest theoretical guarantee; the Chebyshev threshold trades approximately 2–3 percentage points of recall for a provable $P(\text{FP}) \leq 1/k^2$ bound, offering a principled operating point for safety-critical deployments where false alarms carry high operational cost.

5. Discussion

5.1. Interpretation of Results

The ablation results provide strong evidence for our sensor-specific design hypothesis. Each proposed component contributes independently and monotonically, yielding a 40% cumulative MSE reduction without negative interactions. This is not merely the result of adding parameters—the AUAA module alone contributes 25% MSE reduction, substantially exceeding MC dropout's 5% contribution, despite both operating on the same uncertainty estimates. The difference lies in *how* uncertainty is used: MC dropout adds it as an output-level scoring term, while AUAA integrates it into attention computation, actively reshaping the model's internal information flow.

The dynamic weight adapter's 15-percentage-point contribution validates the context-dependence hypothesis. During validation, $\alpha_t \approx 0.494$, indicating a near-equal balance between reconstruction error and uncertainty signals. This balanced operating point suggests that in realistic environments, neither signal alone suffices—the optimal combination is context-dependent, and the adapter learns this dependence end-to-end.

5.2. Comparison with Prior Stochastic TCN Frameworks

Our framework extends stochastic TCNs along three architectural dimensions. First, **uncertainty integration depth**: prior work applies uncertainty post hoc as a scoring term, while our AUAA feeds it into attention computation. Second, **adaptive vs. fixed scoring**: our dynamic weight adapter replaces static α with a context-sensitive mechanism. Third, **explanatory capability**: our feature-importance learner provides native per-sensor attributions absent from prior frameworks. The SWaT cross-dataset improvement (+2.5% F1) confirms that these architectural differences yield measurable gains beyond the IAQ domain.

5.3. Uncertainty Calibration

Qualitatively, MC dropout uncertainty estimates tracked prediction difficulty: uncertainty increased during rapid environmental transitions (morning ventilation onset, occupancy changes) and decreased during stable nighttime periods. The architectural integration of uncertainty distinguishes this work from standalone calibration studies [8,11].

5.4. Sensor-Specific Design Justification

Each architectural choice is motivated by a concrete sensor-characteristic argument:

AUAA justified by sensor heterogeneity. The uncertainty gate $1/(1+\sigma_t)$ is motivated by the observation that sensor noise variance varies both across sensor types (electrochemical VOC vs. thermistor temperature) and over time (sensor aging, environmental interference). A mechanism that uniformly treats all attention positions is suboptimal when some positions carry substantially more reliable information than others.

Dynamic weighting justified by sensor degradation. Fixed-weight scoring implicitly assumes constant sensor quality—an assumption violated by real sensor deployments where CO_2 sensor drift and VOC sensitivity degradation evolve on timescales of days to weeks. The dynamic adapter's GRU state provides memory of recent sensor behavior, enabling adaptation to degradation trends.

Attribution justified by maintenance workflows. In facility management, an anomaly attributed to CO_2 triggers HVAC inspection, while one attributed to $PM_{2.5}$ triggers filter replacement—different sensor attributions lead to different operational responses. The feature-importance learner provides this sensor-level granularity as a native output.

5.5. Design Extensions and Deployment Considerations

The sensor-specific design principles established in this work suggest natural extensions. Training under normal-operation assumptions means contamination by undetected anomalies can bias predictive distributions—robust training strategies such as trimmed loss functions offer a direct mitigation path. The current single-step prediction scope can be extended to multi-horizon forecasting for detecting gradual drift patterns that emerge over longer timescales. For higher-frequency sensor streams (e.g., 100 Hz vibration monitoring), model distillation or reduced MC sample counts preserve the architectural benefits within tighter latency budgets. The fixed sensor set assumption can be relaxed through dynamic input masking, and online learning variants would address concept drift in non-stationary environments.

6. Conclusions

We introduced an uncertainty-calibrated TCN framework that treats predictive uncertainty as an internal architectural signal rather than a post-hoc add-on. The framework is motivated by three sensor-specific challenges—heterogeneous reliability drift across sensor types, context-dependent anomaly scoring requirements, and the need for actionable per-sensor diagnostic output—and integrates three tightly coupled components: (1) the Adaptive Uncertainty-Aware Attention (AUAA) mechanism gates temporal attention by per-sensor predictive uncertainty from MC dropout; (2) the Dynamic Weight Adapter learns context-sensitive blending of reconstruction error and uncertainty; and (3) the feature-importance learner provides per-sensor attributions with Chebyshev-based false-positive-rate guarantees.

Comprehensive experiments on four-month IAQ sensor data demonstrated a 40% cumulative MSE reduction (R^2 0.9953, SMAPE 1.87%) with monotonic ablation gains: AUAA contributes 25% standalone improvement and the dynamic weight adapter adds 15 percentage points. Cross-dataset validation on SWaT achieved F1=0.81 (+2.5% vs. stochastic TCN), confirming domain-agnostic benefits. Per-sensor attribution (Humidity 21.0%, VOC 20.5%, CO_2 20.4%, Dust 19.2%, Temperature 18.9%) provides actionable sensor-level interpretability, while Chebyshev concentration bounds guarantee $P(\text{FP}) \leq 1/k^2$ without distributional assumptions. With sub-50ms inference latency and 15% parameter overhead, the framework is suitable for real-time deployment in smart buildings, environmental monitoring, and industrial process control.

Operational Guarantee: Applying the Chebyshev inequality to anomaly score distributions yields a distribution-free false-positive-rate bound: for any score distribution with finite variance and threshold $\tau = \bar{s} + k \cdot \sigma_s$, the detector satisfies $P(\text{FP}) \leq 1/k^2$. This enables principled threshold selection without parametric distributional assumptions, providing an operational guarantee that complements the empirical percentile-based approach. **Author**

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